Lilly Lecture 1987

The Triumvirate: β-Cell, Muscle, Liver

A Collusion Responsible for NIDDM

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The maintenance of normal glucose homeostasis depends on three simultaneously ongoing processes that must occur in a coordinated fashion (Table 1). After the ingestion of glucose, insulin secretion by the pancreas is stimulated, and the combination of hyperinsulinemia plus hyperglycemia must effectively promote glucose uptake by splanchnic (liver and gut) and peripheral tissues (primarily muscle) and suppress hepatic glucose production (HGP). From this brief overview, it is obvious that defects at the level of the β-cell, muscle, or liver can lead to the development of glucose intolerance or overt diabetes mellitus. In this article, the concept is developed that there are two primary defects responsible for the pathogenesis of non-insulin-dependent diabetes mellitus (NIDDM). In some NIDDM patients the primary defect starts at the level of the β-cell and manifests itself as an impairment in insulin secretion; these individuals are represented by the lean diabetic patient. In other NIDDM patients the primary defect starts as an impairment in tissue (muscle and liver) sensitivity to insulin; these individuals are represented by the obese diabetic. However, whichever defect—diminished insulin secretion or insulin resistance—initiates the development of NIDDM, it will subsequently lead to the emergence of the second abnormality. In fact, both defects must be present simultaneously before moderate to severe glucose intolerance ensues. Although deranged hepatic glucose metabolism plays an important role in maintaining the diabetic state once it has become firmly established, the liver probably does not play a major role in the early development of fasting hyperglycemia in NIDDM patients.

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INSULIN SECRETION

Over the last decade, we have performed oral glucose (100-g) tolerance tests (OGTT) in many normal-weight NIDDM subjects. All subjects had fasting hyperglycemia (mean ± SE 148 ± 8). After glucose ingestion, the plasma glucose concentrations were >200 mg/dl at all time points, thus fulfilling the revised National Diabetes Data Group criteria for the diagnosis of diabetes mellitus (1). Both the fasting plasma insulin concentration and the stimulated plasma insulin levels after the oral glucose load were higher in the NIDDM patients than in age- and weight-matched control subjects. Recent studies have shown that even in NIDDM subjects with more severe fasting hyperglycemia, >200–220 mg/dl, the daily plasma insulin profile is normal (2). Evidence as simple and reproducible as this makes it clear that the glucose intolerance in many NIDDM subjects cannot result solely from an absolute deficiency of insulin. However, the plasma insulin response, although slightly increased in absolute terms, should not be construed to be normal. The β-cell response is largely determined by the prevailing plasma glucose concentration (3,4); thus, when viewed in the context of the marked hyperglycemia, it is clear that there is a relative defect in the ability of the pancreas to secrete insulin in NIDDM.

The literature is replete with studies that use the OGTT and intravenous glucose tolerance test (IVGTT) to examine insulin secretion in NIDDM. Table 2 summarizes the data from 32 studies that have examined insulin secretion in normal-weight NIDDM subjects with fasting hyperglycemia (for a complete listing of the 32 publications see ref. 5). Several points should be emphasized.

First, in the postabsorptive state the plasma insulin concentration has invariably been found to be normal or increased. Furthermore, in studies where normal fasting plasma insulin concentrations have been reported, the levels have uniformly been in the high-normal range. This is not surprising because diabetic subjects have fasting hyperglycemia, and this will present a persistent stimulus for insulin secretion throughout the day. In fact, when basal insulin

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TABLE 1
Factors responsible for maintenance of normal glucose homeostasis in humans

Insulin secretion
Stimulation of glucose uptake
Peripheral tissues (muscle)
Glucose oxidation
Nonoxidative glucose disposal (glycogen synthesis, glycolysis)
Splanchnic tissues (liver plus gut)
Suppression of hepatic glucose production

secretion has been estimated in NIDDM patients by use of C-peptide, it has been found to be increased (6). This basal hyperinsulinemia may be an important factor in the pathogenesis of the insulin resistance that characterizes NIDDM individuals.

Second, only half of the 32 reported series cited in ref. 5 (containing ~500 NIDDM patients) demonstrated a decrease in the absolute amount of insulin secreted in response to oral or intravenous glucose (Table 2). In the other 16 series, the plasma insulin response was either normal (n = 11) or increased (n = 5). These results are similar to our own and again suggest that factors in addition to impaired insulin secretion must contribute to the defect in glucose metabolism. Another important point is that the early phase of insulin secretion during both OGTT (0- to 60-min period) and IVGTT (0–10 min) tends to be reduced in NIDDM subjects with fasting plasma glucose >110–120 mg/dl (7,8; Table 1). This may have important pathogenetic consequences because this early burst of insulin release appears to play an important role in priming the insulin target tissues, primarily the liver, that are responsible for the maintenance of glucose homeostasis (9,10).

From Table 2, it is obvious that the plasma insulin response to both oral and intravenous glucose is quite heterogeneous in NIDDM subjects. The reason for this is shown in Fig. 1. Many studies (11–20), including our own (R.A.D., E. Ferrannini, and D.C. Simonson, unpublished observations), have shown that the relationship between the plasma insulin response and the severity of diabetes is complex. When the fasting plasma glucose concentration, an index of the severity of diabetes, is plotted against the mean plasma insulin response to a 100-g OGTT, a characteristic inverted-U–shaped curve is obtained. An individual with a fasting glucose of 80 mg/dl would be expected to have a mean plasma insulin concentration of ~50 μU/ml during the 2 h after glucose ingestion. As the normal individual starts to become diabetic and the fasting plasma glucose concentration rises, the β-cell recognizes that the glucose homeostatic mechanism has become disrupted, and it augments its insulin secretory capacity in an attempt to overcome the disturbance in glucose metabolism (Fig. 2). Thus, an individual with a fasting glucose of 120 mg/dl (i.e., impaired glucose tolerance or early diabetes mellitus) will secrete approximately twice as much insulin as a normal person with a fasting glucose concentration of 80 mg/dl (Fig. 1). However, once the fasting glucose level exceeds 120 mg/dl, the β-cell can no longer maintain its accelerated rate of insulin secretion, and further increments in the plasma glucose concentration are associated with a progressive decline in insulin secretion. Thus, a diabetic person with a fasting glucose of 150–160 mg/dl will secrete an amount of insulin similar to that in a normal nondiabetic individual (Fig. 1). With further increases in the fasting glucose concentration to >160 mg/dl, the plasma insulin response, when viewed in absolute terms, becomes insulinopenic. Finally, when the basal glucose level exceeds 200–220 mg/dl, the plasma insulin response to a glucose challenge becomes markedly blunted. Nonetheless, the postabsorptive rate of insulin secretion remains elevated, and fasting hyperinsulinemia is characteristic of the diabetic patient (6).

Because this inverted-U–shaped curve relating insulin secretion to the fasting plasma glucose concentration so strongly resembles Starling's curve of the heart, I have called it Starling's curve of the pancreas. In a recent study involving 77 normal-weight NIDDM subjects (R.A.D., E. Ferrannini, and D.C. Simonson, unpublished observations), we have shown that the same inverted-U–shaped curve describes the relationship between the fasting plasma glucose level and the fasting plasma insulin concentration. The preceding synthesis is consistent with the natural history of the development of NIDDM in the rhesus monkey (22). As monkeys age, they become obese, and a high percentage develop typical NIDDM. Longitudinal and cross-sectional studies have documented that the earliest detectable abnormality that predicts the onset of diabetes mellitus is an increase in the fasting and glucose-stimulated plasma insulin concentration

TABLE 2
Summary of plasma insulin response during glucose tolerance tests in nonobese NIDDM subjects with fasting hyperglycemia

	Fasting insulin	Plasma insulin response to glucose		
		Early*	Late	Total
Decreased	0	21	13	16
Normal	27	6	12	11
Increased	5	5	7	5

For sources of the 32 publications cited herein, see ref. 5.
*The early phase of insulin secretion refers to the 0- to 10-min period during the intravenous glucose tolerance test and the 0- to 60-min period during the oral glucose tolerance test.

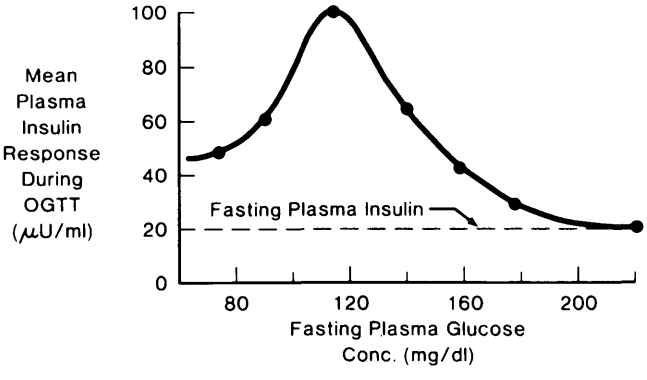


FIG. 1. Starling's curve of pancreas. In normal-weight patients with impaired glucose tolerance and mild diabetes mellitus, plasma insulin response to ingested glucose increases progressively until the fasting glucose concentration reaches 120 mg/dl. Thereafter, further increases in fasting glucose level are associated with progressive decline in insulin secretion. Same curve depicts relationship between fasting plasma insulin and glucose concentration. See text for details.

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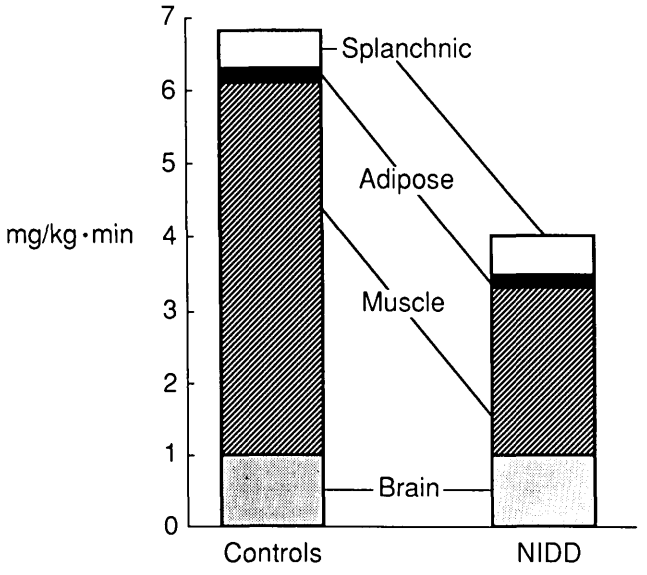


FIG. 2. Summary of glucose metabolism during euglycemic insulin (+100 μU/ml)-clamp studies performed in 38 normal-weight NIDDM subjects and in 33 age-matched controls. See text for more detailed discussion. Data from refs. 5 and 36–39 and unpublished results.

(22). Eventually, however, this high rate of insulin secretion cannot be maintained, and the monkey starts on the downward slope of Starling's curve. At this point, marked fasting hyperglycemia and glucose intolerance ensue. Unfortunately, insulin sensitivity was not measured in this study (22). It is likely that as the monkey grows older and more obese, he becomes insulin resistant and slightly glucose intolerant, although the fasting glucose level does not rise noticeably because of the compensatory hyperinsulinemia.

In summary, the data presented in Table 1 document that the plasma insulin level, when viewed in absolute terms, is normal or increased in most diabetic patients with mild to moderate fasting hyperglycemia. Therefore, insulin resistance must play an important role in the development of the glucose intolerance.

INSULIN SENSITIVITY

Several investigators, using various techniques, have examined insulin action in NIDDM (Table 3). Himsworth and Kerr (23), using the combined OGTT–intravenous insulin tolerance test, were the first to examine tissue sensitivity to insulin in diabetic patients. Subjects consumed a standard oral glucose load while receiving an intravenous bolus of regular insulin. Himsworth and Kerr found that the ability of the injected insulin to promote the disposal of the ingested glucose load was markedly impaired in normal-weight NIDDM subjects. Subsequent investigators, using the insulin tolerance test, also demonstrated a blunted decline in the plasma glucose concentration in NIDDM patients with fasting hyperglycemia (24,25). Similarly, an impairment in insulin action in NIDDM has been demonstrated with the forearm perfusion technique (26,27) and with radioisotope turnover studies (28) and the modified intravenous tolerance test (29).

In 1970, Shen et al. (30) developed a new method to examine insulin sensitivity in vivo. This technique, which has variably been referred to as the quadruple (combined epinephrine, propranolol, insulin, and glucose) infusion technique or the insulin suppression test, has been used extensively to assess insulin sensitivity in NIDDM, and the results have uniformly documented the presence of moderate to severe insulin resistance (30–32). Subsequently, several groups substituted somatostatin for epinephrine and propranolol in the quadruple-infusion cocktail to exlude any potential antagonistic effects of epinephrine on hepatic and/or peripheral tissues (33,34). These investigators also found higher steady-state plasma glucose levels in their NIDDM patients, again indicating an impairment in tissue sensitivity to insulin.

The most conclusive documentation of insulin resistance in NIDDM has come from studies that have employed the more physiologic insulin-clamp technique (35). In the largest series involving normal-weight diabetic patients, we reported the results of euglycemic insulin-clamp studies in 38 lean NIDDM and 33 age- and weight-matched control subjects (5,36–39). The mean fasting glucose concentration in the diabetic group was 150 ± 8 mg/dl, representing a mild to moderate degree of fasting hyperglycemia. None of the diabetic subjects was taking any medications for at least 2 wk before study, and none had ever received insulin. Diabetic patients with more severe degrees of glucose intolerance were not included. This study focused on the early stages of evolution of NIDDM. Patients with severe diabetes (fasting plasma glucose concn >200 mg/dl) are known to be insulinopenic (Fig. 1), and insulin deficiency is associated with the emergence of several intracellular defects in insulin action (40,41). In addition, suppression of HGP by insulin becomes impaired in diabetic individuals with fasting glucose levels >200 mg/dl.

All of the diabetic and control subjects received a euglycemic insulin clamp in combination with [^{3}H]glucose to quantitate rates of HGP. With the insulin-clamp technique the fasting plasma insulin concentration is acutely raised and maintained at 100 μU/ml above the basal level by a primed continuous infusion of insulin (35). To prevent hypoglycemia with its attendant neuroendocrine counterregulatory response, an exogenous glucose infusion is administered at a rate designed to clamp the plasma glucose concentration at euglycemic (100-mg/dl) levels. Under these steady-state conditions of constant glycemia, the amount of glucose infused must be taken up by tissues in the body and provides a measure of insulin-mediated glucose metabolism.

The total amount of glucose metabolized by the entire body during the insulin-clamp study in the NIDDM and control groups is shown in Fig. 2 (5,36–39). The mean rate of

TABLE 3
Techniques demonstrating insulin resistance in NIDDM

Technique	Refs.
Combined glucose and intravenous insulin tolerance tests	23
Intravenous insulin tolerance test	24,25
Forearm perfusion technique	27
Insulin suppression test	30–32
Somatostatin modification of insulin-suppression test	33,34
Radioisotope turnover studies	28
Insulin-clamp technique	35–39,48
Modified intravenous glucose tolerance test	7

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insulin-mediated glucose metabolism was reduced by 35–40% in NIDDM compared with control subjects. Furthermore, even the most sensitive diabetic patient fell at least 1SD below the control subjects. Thus, not only was there a severe degree of insulin resistance in the NIDDM group, but there was a complete separation between the diabetic and control subjects.

SITE OF INSULIN RESISTANCE

Insulin controls glucose homeostasis through three coordinated mechanisms (Table 1): suppression of HGP, stimulation of glucose uptake by the splanchnic (hepatic plus gastrointestinal) tissues, and stimulation of glucose uptake by peripheral tissues. Glucose uptake, in turn, depends on two major metabolic pathways: glucose oxidation and glucose storage (or nonoxidative glucose disposal). Each of these processes may be a cause of the insulin resistance in NIDDM.

HGP. During all insulin-clamp studies, [3-^{3}H]glucose was routinely used to quantitate HGP in the basal state and the suppression of HGP in response to hyperinsulinemia (5,36–39). In the basal postabsorptive state the liver of healthy subjects produces glucose at the rate of ~1.8–2.2 mg · kg^{-1} · min^{-1}. This glucose flux is critical to meet the obligatory need of the brain for glucose during interabsorptive periods. After feeding, on the other hand, it is critical that the insulin released into the portal vein bind to specific receptors on the hepatocyte and suppress HGP. If this does not occur, there will be two inputs of glucose into the body (the liver and the ingested glucose from the gastrointestinal tract), and marked hyperglycemia will develop.

In the NIDDM patients in the postabsorptive state there was a small but significant ($P < .001$) increase in HGP of ~0.5 mg · kg^{-1} · min^{-1}. Although this difference may seem small, it is impressive to calculate that over 24 h, the liver of a 70-kg diabetic individual will add an additional 50 g of glucose to the system circulation. Furthermore, in NIDDM subjects the increase in basal HGP was closely correlated ($r = .920$, $P < .001$) with the degree of fasting hyperglycemia (5,36–39). These data indicate that an excessive rate of HGP is a major factor responsible for the elevated fasting plasma glucose concentration in NIDDM individuals with overt fasting hyperglycemia (>140 mg/dl).

Two additional points concerning the elevated rates of basal HGP should be emphasized. First, the fasting plasma insulin concentration in the NIDDM subjects was twice that in control subjects. Because hyperinsulinemia is a potent inhibitor of HGP (42,43), significant hepatic resistance to the action of insulin must be present in the postabsorptive state to explain the increased output of glucose by the liver. Second, the fasting plasma glucose concentration was significantly increased in the diabetic group. Because hyperglycemia per se has a suppressive action on HGP (42,44,45), there may also be glucose resistance with respect to the inhibitory effect of hyperglycemia on HGP.

Although HGP is elevated in the postabsorptive state (implying a significant degree of hepatic insulin resistance), in response to an increment in plasma insulin concentration (100 μU/ml) above baseline (i.e., during the insulin-clamp study), there was >90% suppression of HGP in the diabetic group, and this result was essentially identical to that observed in the control group. Thus, under these experimental conditions, the site of insulin resistance during the insulin clamp is not at the level of the liver, at least with regard to the suppression of HGP. A normal suppressive action of insulin on HGP in NIDDM subjects has also been reported by others (46,47). Note, however, that in NIDDM patients with more severe fasting hyperglycemia (>200–220 mg/dl), an impairment in the ability of insulin to suppress HGP has been demonstrated (48). Thus, as the diabetes becomes more severe, a defect in the inhibitory effect of insulin on HGP may become more prominent. We have also shown that at lower plasma insulin levels, a defective suppression of HGP may manifest itself even in less hyperglycemic patients; i.e., the dose-response curve relating HGP to the plasma insulin concentration is shifted to the right (49).

The demonstration that an accelerated rate of HGP contributes to fasting hyperglycemia in NIDDM subjects has important therapeutic implications. If a poorly controlled NIDDM patient has failed on the standard regimen of diet, exercise, and sulfonylureas and insulin therapy must be initiated, at what time of the day should insulin be administered and what type of insulin should be given? Our results (5,37–39), as well as those of others (11,50–54), have shown that HGP correlates closely with fasting hyperglycemia. This excessive HGP occurs primarily at night, when the patient has gone to sleep and the plasma insulin concentration has declined to basal levels. It makes theoretical sense, therefore, to administer a long-acting insulin preparation in the evening before the individual goes to bed. This will ensure some level of hyperinsulinemia throughout the sleeping hours, hepatic overproduction of glucose will be restrained, and the fasting glucose concentration will be maintained within the normal range. Some experimental validation of the usefulness of bedtime insulin has been provided by Riddle et al. (55).

It also makes sense to develop new insulin preparations that are hepatic specific (56). The efficacy of a hepatic-specific insulin was first shown in diabetic dogs by Spangler (57), who packaged insulin in liposomes containing an asialoglycoprotein-receptor ligand, digalactosyl diglyceride in the membrane of the lipid droplets. Because receptors for asialoglycoprotein are located only in the liver, all of the infused liposomes and, therefore, all of the infused insulin were cleared by the liver, HGP declined to basal levels, and normoglycemia was restored (57). Unfortunately, liposomes must be injected intravenously, so they do not represent a practical means of treating NIDDM patients. However, Spangler and Thornton (58) have biochemically altered the insulin molecule by conjugating lactobionic acid to the NH_2-terminals of the A- and B-chains, and this insulin preparation has been shown to be highly hepatic specific. Most recently, proinsulin has been shown to have relative hepatic specificity (compared with its stimulatory effect on peripheral tissues), but further studies are needed to demonstrate its clinical efficacy as a hepatic-specific insulin (59).

Glucose, which is released by the liver, can be derived either from glycogenolysis or from gluconeogenesis (44,45). Studies by Waldhausl et al. (60) have shown that hepatic gluconeogenic precursor, especially lactate, uptake is increased in NIDDM subjects. Consistent with this observation, Consoli et al. (61), using radioisotope turnover techniques, have shown that the increase in HGP above baseline can

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be accounted for primarily by accelerated gluconeogenesis. If this is correct, then it makes sense to develop new oral antidiabetic agents that have the ability to inhibit hepatic gluconeogenesis as their primary mechanism of action. A drug that has potential in this regard is dichloroacetate. When administered to NIDDM patients, a pronounced blood glucose–lowering effect has been described (62). Although dichloroacetate is known to enhance glucose metabolism by peripheral tissues (63), it also is a powerful inhibitor of gluconeogenesis through its direct effect on the liver and by altering the delivery of gluconeogenic substrates (63,64).

Splanchnic (hepatic) glucose uptake. Results from [3-^{3}H]glucose kinetic analysis demonstrate that the insulin resistance observed during the euglycemic insulin clamp cannot be explained by an impaired suppression of HGP. A second potential mechanism that might account for the impairment in insulin action during the insulin clamp is a decrease in glucose uptake by the liver. It is difficult to catheterize the portal vein in humans, and therefore it is not possible to directly examine glucose disposal by the liver. However, by use of the hepatic vein catheterization technique in combination with the euglycemic insulin clamp, we have examined the contribution of the splanchnic tissues to overall glucose homeostasis in 10 NIDDM subjects with mild to moderate fasting hyperglycemia (39). The hepatic vein catheter technique allows the estimation of the net amount of glucose taken up by the splanchnic region from the product of hepatic blood flow and the arterial-hepatic venous blood glucose concentration gradient. Note that the data obtained from hepatic vein catheterization represent net glucose uptake by the entire splanchnic area, including hepatic and gastrointestinal tissues.

A net release of glucose from the splanchnic area (i.e., negative balance) occurred in the basal state in both control and NIDDM subjects (Fig. 3), reflecting glucose production by the liver. A prompt suppression of splanchnic glucose output occurred after insulin administration (reflecting the inhibition of HGP by insulin), and by 20 min the net balance of glucose across the splanchnic bed was zero (i.e., there was no net uptake or release). After 2 h of sustained hyperinsulinemia, there was a small net uptake of glucose by the splanchnic area (i.e., positive balance), which averaged ~0.5 mg · kg^{-1} · min^{-1}. There was no difference between diabetic and control subjects in the amount of glucose that was taken up by the splanchnic tissues. Thus, a defect in splanchnic (hepatic) glucose uptake could not be demonstrated in NIDDM.

Figure 3 also illustrates another important point, i.e., that under conditions of euglycemic hyperinsulinemia, very little of the infused glucose is taken up by the splanchnic (and therefore hepatic) tissues. During the insulin clamp, the rate of whole-body glucose uptake averaged 7 mg · kg^{-1} · min^{-1}, and of this, only 0.5 mg · kg^{-1} · min^{-1} (7%) was removed by the splanchnic region. Because the difference in total-body glucose uptake between the NIDDM and control groups during the euglycemic insulin-clamp study was 2.5 mg · kg^{-1} · min^{-1}, it is clear from a purely quantitative standpoint that a defect in splanchnic (hepatic) glucose removal could never account for the magnitude of impairment in total-body glucose uptake.

The net splanchnic glucose balance is the sum of two

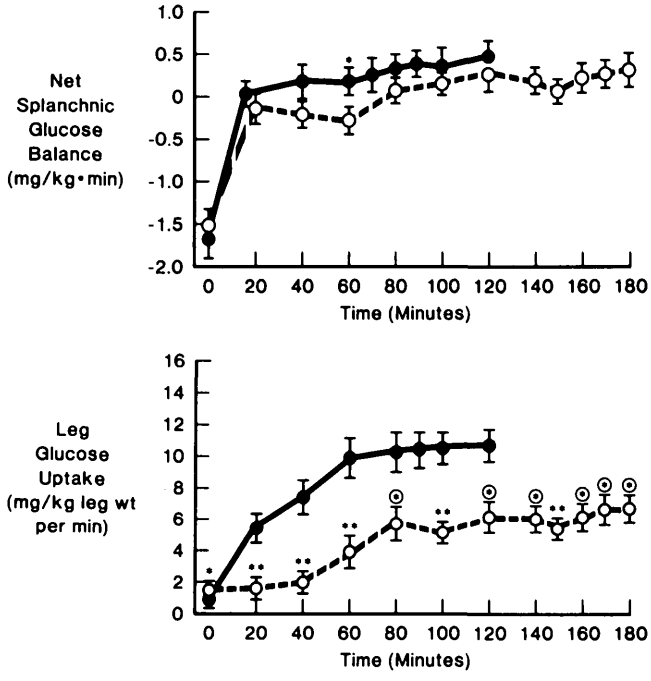


FIG. 3. ***Top:*** **time course of change in net splanchnic glucose balance in NIDDM (○) and control (●) subjects. Difference between diabetic and control subjects is small, statistically insignificant, and cannot account for marked impairment in total-body glucose metabolism observed during euglycemic insulin-clamp study. Note that total amount of glucose disposed of by splanchnic area represents <10% of infused glucose load in both groups.** ***Bottom:*** **time course of change in leg glucose uptake in NIDDM (○) and control (●) subjects. In postabsorptive state, glucose uptake in diabetic group was significantly greater than in control subjects. However, ability of insulin (euglycemic insulin clamp) to stimulate leg glucose uptake was reduced by 50% in diabetic subjects. *P < .05 vs. control; **P < .001 vs. control; ⊙P < .01 vs. control. (Reproduced from** ***J Clin Invest*** **76:149–55, 1985, by copyright permission of the American Society for Clinical Investigation.)**

simultaneously ongoing processes: splanchnic glucose uptake and HGP. Because HGP was calculated independently with [^{3}H]glucose, the actual uptake of glucose by the splanchnic area could be calculated from the difference between the HGP ([^{3}H]glucose) and the net splanchnic glucose balance (hepatic vein catheter). In the basal state, splanchnic glucose uptake averaged 0.5 ± 0.1 mg · kg^{-1} · min^{-1}, a value identical to that observed during the insulin clamp. These results indicate that the liver and extrahepatic splanchnic tissues are insensitive to the effect of insulin to augment glucose uptake (39,42,44,45). Thus, it is clear that the insulin resistance observed during an insulin-clamp study can never be explained by a decrease in glucose disposal by the splanchnic (hepatic) tissues.

In summary, neither a defect in suppression of HGP nor a defect in hepatic glucose uptake can account for the impairment in tissue sensitivity to insulin during the euglycemic insulin-clamp study in NIDDM individuals with modest fasting hyperglycemia (<200 mg/dl).

Peripheral (muscle) glucose uptake. Because neither suppression of HGP nor splanchnic glucose uptake is impaired in NIDDM subjects during the insulin clamp, peripheral tissues must be the primary site of insulin resistance. To examine this directly, the same 10 subjects who had the insulin-clamp study performed in combination with hepatic

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venous catheterization also had a catheter inserted into the femoral vein to quantitate glucose exchange across the leg tissues (39; Fig. 3). In the basal state, leg glucose uptake was slightly, although significantly ($P < .01$), increased in the diabetic group (1.8 vs. 1.0 mg · kg^{-1} leg wt · min^{-1}). This increase in tissue glucose uptake is consistent with radioisotope turnover studies, which demonstrate a significant enhancement in the rate of tissue glucose disposal in NIDDM subjects in the postabsorptive state (5,11,36–39,50–54). Note, however, that the metabolic rate of the glucose that is taken up in increased amounts by peripheral tissues, primarily muscle, is not normal in NIDDM patients. Because glucose oxidation (65) and glycogen formation (66) are impaired in NIDDM subjects, an excessive amount of glucose is converted to lactate (39), which is subsequently released and can serve as a substrate (60) to drive gluconeogenesis by the liver (61). This accelerated Cori cycle activity provides an important mechanism that sustains the accelerated rate of HGP in diabetic individuals with well-established fasting hyperglycemia. The increased glucose uptake by peripheral tissues is due to the mass action effect of hyperglycemia, which passively drives glucose into cells (44,45).

During the insulin-clamp study in the control subjects, there was a prompt stimulation of leg glucose uptake, which reached a plateau value of ~10 mg · kg^{-1} leg wt · min^{-1} during the last hour of the study (Fig. 3). In contrast, in the diabetic subjects, the ability of insulin to stimulate leg glucose uptake was delayed and markedly blunted. During the last hour of the study, the rate of glucose uptake was reduced by 50% in the diabetic group. These results clearly indicate that the primary site of insulin resistance during the euglycemic insulin clamp in NIDDM subjects resides in peripheral tissues. With the forearm perfusion technique, Kalant et al. (67) and Butterfield and Whichelow (26) have also demonstrated a decreased rate of glucose uptake by peripheral tissues in response to insulin. In humans, adipocytes are inert and respond poorly to the stimulatory effect of insulin on glucose metabolism. Björntorp et al. (68,69) have shown that <1% of an infused or ingested glucose load is taken up by adipose tissue. Thus, the primary leg and forearm tissue responsible for glucose uptake must be muscle. If all muscle tissue in the body responds similarly to leg muscle, most (>90%) of the impairment in total-body glucose uptake during the insulin-clamp study in diabetic subjects would be accounted for (39).

Whole-body glucose economy during the insulin clamp is summarized schematically in Fig. 2. The height of each bar represents the total amount of glucose taken up during the insulin clamp in control and NIDDM subjects. Net splanchnic glucose uptake (quantitated by the hepatic venous catheterization technique) is similar in both groups and averages 0.5 mg · kg^{-1} · min^{-1}. Adipose tissue glucose uptake, which was not directly measured, is assumed to represent 1% of total glucose disposal (68,69). Brain glucose uptake, estimated to be 1.0–1.2 mg · kg^{-1} · min^{-1} in the postabsorptive state (70,71), is assumed not to have been affected by hyperinsulinemia. In control subjects, muscle glucose uptake (extrapolated from leg catheterization data) accounts for ~75% of the total glucose uptake. In NIDDM subjects, it appears that the largest part, if not all, of the impairment in insulin-mediated glucose uptake can be accounted for by a defect in muscle glucose disposal.

It could be argued that results obtained during a euglycemic insulin-clamp study might not be representative of those obtained after glucose ingestion. To examine this question 12 normal-weight NIDDM (fasting plasma glucose 189 ± 19 mg/dl) and 10 age- and weight-matched control subjects ingested 1 g/kg (70 g total) of [1-^{14}C]glucose, and the endogenous glucose pool was labeled with [3-^{3}H]glucose (72). With this double tracer technique, we were able to quantitate both tissue glucose uptake and suppression of HGP. Glucose tolerance during the 3.5-h postglucose ingestion was markedly impaired in the diabetic group, despite a plasma insulin response that was similar to that in the control group. Tissue glucose uptake was significantly reduced in diabetic compared with control subjects (44 vs. 60 g over 3.5 h, $P < .05$); during the 3.5 h after glucose ingestion, HGP continued at a higher rate in the diabetic than in the control group (17 vs. 10 g over 3.5 h), indicating resistance to the inhibitory effects of insulin and hyperglycemia on HGP. In summary, both diminished tissue glucose uptake and impaired suppression of HGP contribute to the glucose intolerance after glucose ingestion. From the quantitative standpoint, however, the defect in tissue glucose disposal is twice that for suppression of HGP. Nearly identical results have been reported recently by Firth et al. (73).

SUMMARY OF INSULIN RESISTANCE

The above results firmly establish that tissue insensitivity to insulin is an important pathogenetic disturbance that contributes to the glucose intolerance of NIDDM subjects. However, it must be remembered that the tissues responsible for the insulin resistance in the basal state are quite different from those responsible for the insulin resistance in the insulin-stimulated state. This distinction has important implications for understanding the pathophysiology, as well as the therapy, of NIDDM.

In the basal state the liver represents a major site of insulin resistance. This is reflected by an overproduction of glucose despite the presence of fasting hyperinsulinemia and hyperglycemia. The increase in HGP is a key factor responsible for fasting hyperglycemia in the NIDDM individual. Although tissue glucose uptake in the postabsorptive state is increased when viewed in absolute terms, the efficiency with which glucose is cleared by muscle is diminished. In contrast, after glucose infusion or ingestion, i.e., in the insulin-stimulated state, muscle is the primary tissue responsible for the insulin resistance. This impairment in insulin-mediated muscle glucose uptake accounts for the defect in glucose removal after carbohydrate administration. In diabetic subjects with more severe fasting hyperglycemia, impaired suppression of HGP after glucose ingestion also contributes to the glucose intolerance. In designing a therapeutic regimen for NIDDM patients, it is important to introduce interventions that will *1*) inhibit basal HGP, thereby lowering the fasting plasma glucose concentration, and *2*) stimulate muscle glucose uptake, thus promoting the disposal of an ingested or infused glucose load.

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MUSCLE VERSUS LIVER: WHICH IS RESPONSIBLE FOR FASTING HYPERGLYCEMIA IN EARLIEST STAGES OF NIDDM?

From the preceding discussion it is clear that once overt fasting hyperglycemia (>140 mg/dl) has developed, HGP is elevated and correlates well with the elevated basal glucose concentration (5,36–39). However, by the time the fasting glucose has reached 140 mg/dl, the diabetic individual has had the disease for a long time. Thus, these studies cannot answer the question as to whether, in the earliest stages of NIDDM, fasting hyperglycemia results from an excessive rate of HGP or from a decreased efficiency of glucose uptake by peripheral tissues, primarily muscle. To address this issue, we performed [^{3}H]glucose turnover studies in the postabsorptive state in 77 NIDDM patients who were within 25% of ideal body weight (based on medium-framed individuals from Metropolitan Life Insurance tables, 1959; R.A.D., E. Ferrannini, and D.C. Simonson, unpublished observations). The mean ideal body weight was 110%. Seventy-two age-, sex-, and weight-matched healthy subjects served as controls. The mean duration of diabetes was 7 ± 1 yr. Seventeen of the diabetic subjects were treated with diet alone, and 60 were taking an oral hypoglycemic drug, which was stopped at least 7 days before the study. Thirty-three of the NIDDM subjects had fasting plasma glucose levels <140 mg/dl, whereas 44 had levels >140 mg/dl. All of the diabetic subjects had an abnormal OGTT (1).

If only the 33 diabetic individuals with fasting plasma glucose concentrations <140 mg/dl are examined, it is clear that the mean rate of HGP (1.85 ± 0.03 mg · kg^{-1} · min^{-1}) is virtually identical to the control group (1.84 ± 0.02 mg · kg^{-1} · min^{-1}) (Fig. 4, *shaded area*). These results demonstrate that an accelerated rate of HGP cannot explain fasting hyperglycemia early in the course of NIDDM. Note, however, that the regulation of hepatic glucose metabolism is not normal in this group of diabetic patients with mild fasting hyperglycemia. Thus, the basal plasma insulin concentration was nearly twofold greater in the diabetic patients (20 ± 2 vs. 11 ± 1 μU/ml, $P < .001$). A normal rate of HGP in the presence of fasting hyperglycemia demonstrates that hepatic resistance to the suppressive action of insulin is already well established early in the course of NIDDM.

Because excessive glucose production by the liver cannot account for fasting hyperglycemia early in the course of NIDDM, there must be a decrease in the efficiency of tissue glucose uptake. Because the rate of HGP must equal the rate of tissue glucose uptake in the postabsorptive state, it follows that glucose disposal, when viewed in absolute terms, is normal. However, when viewed in context of the increased plasma glucose and insulin levels, it is clear that the efficiency with which glucose is taken up by the diabetic individual's tissues must be reduced early in the course of NIDDM. This is best appreciated by plotting the metabolic clearance rate of glucose against the fasting plasma glucose concentration in diabetic individuals with mild fasting hyperglycemia. Figure 5 (*shaded area*) shows that the glucose clearance rate declines in a steeply progressive and linear ($r = .700$, $P < .001$) fashion as the plasma glucose concentration rises from 105 to 140 mg/dl, whereas no change in HGP is evident. Some (74,75) but not all (43,76) investigators have objected to the use of glucose clearance to provide a quantitative measure of tissue glucose disposal. However, the controversy revolves around whether glucose uptake increases in direct proportion to the increase in plasma glucose concentration. Some increase in the absolute rate of tissue glucose disposal should occur as the plasma glucose concentration increases (43,74–76). The failure to observe any increase in the basal rate of glucose uptake in the diabetic group, despite the presence of fasting hyperglycemia and fasting hyperinsulinemia, indicates that

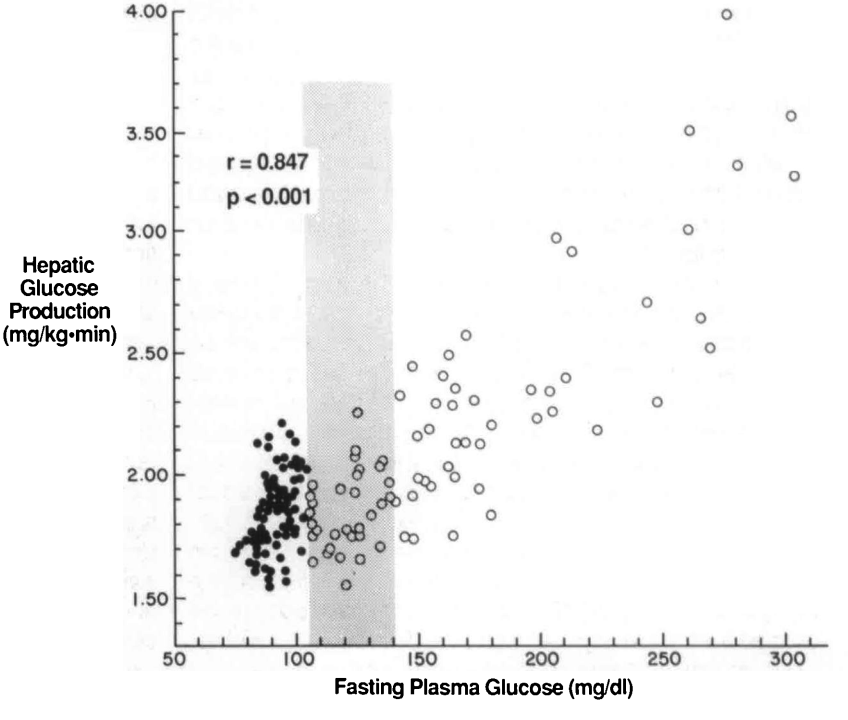


FIG. 4. Summary of hepatic glucose production in 77 normal-weight NIDDM subjects (○) with fasting plasma glucose concentrations ranging from 105 to >300 mg/dl. ●, Age- and weight-matched control subjects ($n = 73$). In 33 diabetic subjects with fasting plasma glucose concentrations <140 mg/dl (*shaded area*), mean rate of hepatic glucose production was virtually identical to controls. In diabetic subjects with fasting plasma glucose levels >140 mg/dl, there was progressive rise in hepatic glucose production that correlated closely ($r = .847$, $P < .001$) with fasting glucose concentration. (R.A.D., E. Ferrannini, and D.C. Simonson, unpublished observations.)

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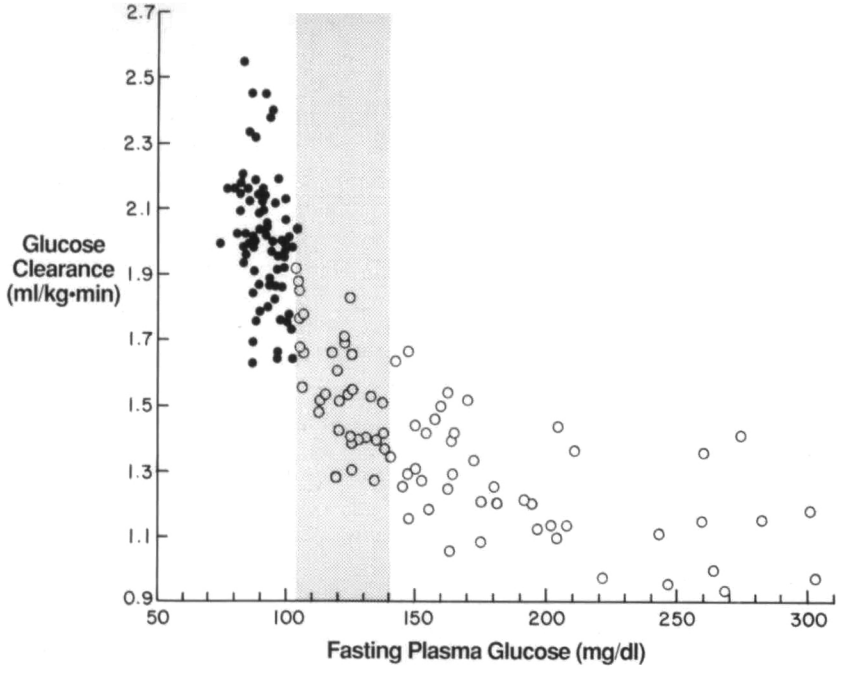


FIG. 5. Summary of metabolic clearance rate of glucose in same 77 normal-weight NIDDM subjects (○) shown in Fig. 4. ●, Age- and weight-matched control subjects (n = 73). In 33 diabetic subjects with fasting plasma glucose concentrations <140 mg/dl (*shaded area*), glucose clearance fell precipitously and was strongly and inversely correlated (r = −.697, P < .001) with increase in plasma glucose concentration. At fasting glucose levels >140 mg/dl, rate of decline in glucose clearance rate began to slow and reached plateau at glucose levels >180 mg/dl. (R.A.D., E. Ferrannini, and D.C. Simonson, unpublished observations.)

one of the earliest discernible abnormalities in NIDDM is a reduction in the glucose clearance or the efficiency at which tissues take up glucose.

The situation with HGP and tissue glucose clearance reverses itself in NIDDM individuals with fasting glucose levels >140 mg/dl. At this level of glycemia, the restraining effect of hyperinsulinemia on the liver is lost, and HGP increases progressively and is directly correlated (r = .847, P < .001) with the rise in fasting glucose concentration (Fig. 4). In contrast, the decline in whole-body glucose clearance tended to plateau at glucose levels of 140–180 mg/dl and changed little at fasting plasma glucose concentrations of 180–300 mg/dl (Fig. 5); i.e., at glucose levels >180 mg/dl the increase in whole-body glucose disposal rate rose in proportion to the increase in basal glucose concentration such that the glucose clearance remained constant, although reduced by ~40–50% compared with the postabsorptive state. This linear increase in glucose disposal at plasma glucose concentrations >180 mg/dl can be viewed as a protective mechanism that prevents the further development of severe hyperglycemia and associated intracellular dehydration.

It is reasonable to ask why the development of fasting hyperinsulinemia in diabetic individuals with mild fasting hyperglycemia is sufficient to prevent overproduction of glucose by the liver but fails to maintain a normal basal rate of glucose clearance. This apparent paradox is best understood by examining the dose-response relationship between the plasma insulin concentration and hepatic glucose disposal (49, R.C. Bonadonna, L. Groop, E. Ferrannini, S. DelPrato, and R.A.D., unpublished observations). When insulin is infused at rates of 0.1 and 0.25 mU · kg^{-1} · min^{-1} in normal individuals to increase the basal (8 μU/ml) insulin concentrations to 15 and 27 μU/ml, basal HGP is suppressed by 33 and 68%, respectively, whereas whole-body glucose uptake fails to increase above baseline. These results indicate that suppression of HGP, in contrast to the stimulation of tissue glucose uptake, is very sensitive to small increments in plasma glucose concentration. Thus, as the plasma glucose concentration increases, for whatever reason, insulin secretion is stimulated (5,6), and the resultant hyperinsulinemia is sufficient to offset any hepatic insulin resistance and maintain a normal absolute rate of basal HGP. The situation with respect to glucose uptake is different. Because small increments in plasma insulin levels do not augment glucose uptake (49,77–79), the development of impaired tissue glucose disposal in NIDDM subjects is not counteracted by the fasting hyperinsulinemia, and the glucose clearance drops progressively.

In conclusion, the results herein indicate that in NIDDM subjects with mild fasting hyperglycemia (<140 mg/dl), the principal disturbance responsible for the elevated postabsorptive plasma glucose concentration is a defect in the efficiency of tissue glucose uptake (Fig. 5). HGP, when viewed in absolute terms, is not increased. In contrast, in NIDDM subjects with more severe fasting hyperglycemia (>140 mg/dl), an augmented rate of HGP is the major factor responsible for the progressive rise in fasting plasma glucose concentration (Fig. 4). The decline in whole-body glucose clearance plateaus at a fasting glucose concentration >160–180 mg/dl and does not contribute further to the progressive rise in fasting glucose concentration observed in more severely diabetic individuals.

DYNAMIC INTERACTION BETWEEN INSULIN ACTION AND INSULIN SECRETION

In the most general sense, an impairment in glucose tolerance could result from either a defect in insulin secretion or tissue resistance to the action of insulin. Unfortunately, no previous study has reported simultaneous measurements of insulin secretion and insulin sensitivity in a sufficient number of diabetic and control individuals to draw meaningful conclusions concerning the interaction between these two variables in the pathogenesis of NIDDM. Following are studies that help to clarify this important issue (80).

Six groups were studied. They were subdivided according

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TABLE 4
Patient characteristics

	n	Age (yr)	Ideal body weight (%)
Obese nondiabetic	47	33	148
Normal glucose tolerance			
Impaired glucose tolerance			
Obese diabetic	35	53	147
Hyperinsulinemic			
Hypoinsulinemic			
Normal-weight diabetic	16	54	109
Control	13	53	100

See text for description of experimental protocol.

to their plasma glucose and insulin responses during a 100-g OGTT (Table 4). Four groups of obese subjects were subdivided: *1*) 24 obese subjects with normal glucose tolerance and 23 obese subjects with impaired glucose tolerance (1); *2*) 35 obese subjects with frank diabetes mellitus, further subdivided into those with a hyperinsulinemic response and those with a hypoinsulinemic response greater or less than, respectively, 1SD of normal-weight control subjects during OGTT (all of the obese subjects were similar in ideal body weight); *3*) 16 normal-weight diabetic subjects; and *4*) 13 normal-weight control subjects matched in age to the normal-weight diabetic group. All subjects ingested 100 g of glucose to provide a measure of glucose tolerance and an index of their insulin secretory capacity. A euglycemic insulin clamp was performed to quantitate whole-body tissue sensitivity to insulin. All insulin-clamp studies were performed with indirect calorimetry to quantitate rates of glucose oxidation and nonoxidative glucose disposal or glucose storage (81); this measure is felt to primarily represent glycogen formation (82–84).

Normal-weight NIDDM subjects manifested a 40–50% decrease in the rate of insulin-mediated whole-body glucose uptake (Fig. 6). This impairment in insulin action results from defects in both the oxidative and nonoxidative pathways of glucose metabolism. How does the degree of insulin resistance in a normal-weight diabetic individual compare with that observed in a nondiabetic obese person? As shown in Fig. 6, obesity per se confers a degree of insulin resistance that is virtually identical to that in the normal-weight diabetic subject. Furthermore, when the contributions of impaired glucose oxidation and glucose storage to the insulin resistance are examined, they appear nearly identical (Fig. 6). Similar results concerning whole-body insulin sensitivity have been reported by Reaven and colleagues (85,86). Despite nearly identical degrees of insulin resistance, the diabetic subjects had fasting hyperglycemia and glucose intolerance, whereas obese nondiabetic individuals had normal oral glucose tolerance. What is the explanation for this marked difference in glucose tolerance between the two groups? The answer relates to the plasma insulin response during the OGTT. Compared with control subjects, the obese nondiabetic group secreted approximately twice as much insulin and thus offset the tissue insensitivity to the hormone. In contrast, the pancreas of the normal-weight diabetic individual, when faced with the same challenge, could not augment its insulin secretory capacity sufficiently to compensate for the insulin resistance. The result was a frankly diabetic state with fasting hyperglycemia and glucose intolerance. It also is possible that activation of insulin-stimulated glucose disposal may be slower and deactivation faster in diabetic individuals (87). This possibility should be further explored.

What happens to insulin sensitivity when obesity and diabetes coexist in the same individual? A precipitous decline in insulin-mediated glucose disposal might be expected. On the contrary, the severity of insulin resistance was only slightly greater in both the hyperinsulinemic and hypoinsulinemic groups than in either the normal-weight diabetic or nondiabetic obese groups (Fig. 6). Likewise, the magnitude of the defects in glucose oxidation and nonoxidative glucose disposal were similar in all four (obese nondiabetic, obese diabetic hyperinsulinemic, obese diabetic hypoinsulinemic, and normal-weight diabetic) (Fig. 6) groups. Although obese diabetic hyperinsulinemic and hypoinsulinemic groups are

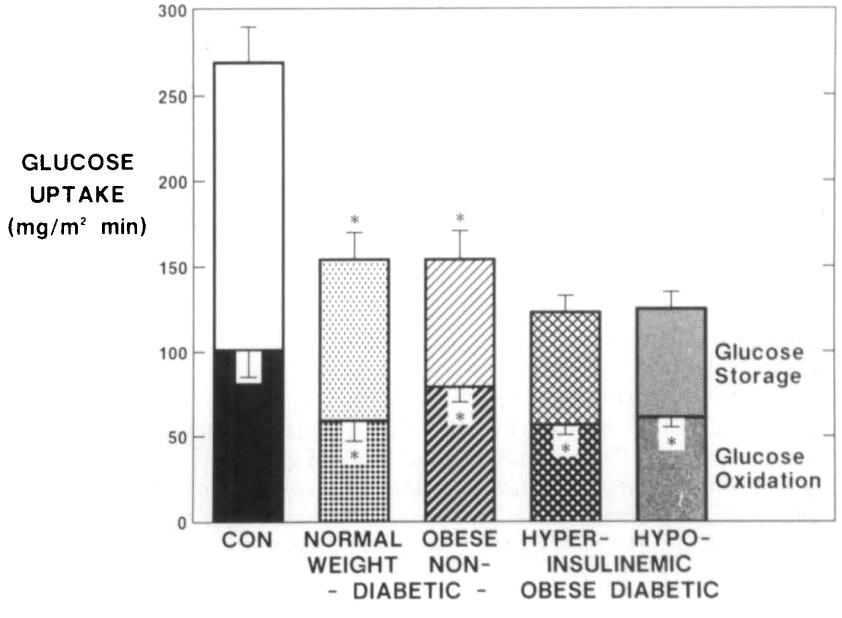


FIG. 6. Insulin-mediated rates (euglycemic insulin-clamp technique) of whole-body glucose uptake (total height of *bar*), glucose oxidation, and nonoxidative glucose disposal in control, normal-weight diabetic, obese nondiabetic, obese glucose-intolerant, and obese diabetic subjects. Obese diabetic patients were subdivided into hyperinsulinemic and hypoinsulinemic groups based on their plasma insulin response during 100-g oral glucose tolerance test. *$P < .001$ vs. control. (Data from ref. 80.)

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equally insulin resistant, the severity of glucose intolerance was significantly worse in the obese hypoinsulinemic group. The reason for the marked impairment in glucose tolerance in the obese diabetic hyposinsulinemic group is related to the severe degree of insulin deficiency. Thus, the obese diabetic group, which is able to mount a hyperinsulinemic response, maintains a better degree of glucose intolerance than the obese insulinopenic group.

The data discussed in the preceding paragraphs are summarized in Fig. 7, which relates *1*) the mean plasma glucose concentration during the OGTT (*top*), *2*) the insulin secretory response during the OGTT (*top*), and *3*) the tissue sensitivity to insulin (*bottom*). In the obese nondiabetic individual there was a dramatic decline in tissue sensitivity to insulin. However, glucose tolerance remained perfectly normal because the β-cell was able to appropriately augment its insulin secretory capacity. As the obese individual became mildly glucose intolerant, there was a further reduction in insulin-mediated glucose disposal. However, there was only a small additional impairment in glucose tolerance since the β-cell was able to further augment insulin secretion to largely offset the worsening insulin sensitivity (Fig. 7). What happens when this obese glucose-intolerant person progresses to overt diabetes mellitus? This is depicted by the obese diabetic group with the hyperinsulinemic response (Fig. 7). There was a marked decline in the OGTT. This deterioration was not due to a worsening of the insulin resistance; rather, the obese diabetic has tipped over the top of Starling's curve and is now on the descending portion (Fig. 1). Even though the plasma insulin response was increased compared with the control group, it was not appropriately elevated for the degree of insulin resistance. In the normal-weight diabetic group, there was a greater demise in glucose tolerance. Again, this was not the result of further deterioration in insulin sensitivity but rather of a greater impairment in insulin secretion. In the obese diabetic with low insulin response, glucose tolerance was most severely impaired due to the marked deficiency in insulin secretion.

The above sequence of events can be summarized as follows. Insulin resistance is an early and universal finding in obesity and in normal-weight NIDDM subjects. Quantitatively, insulin resistance is equally severe in obese and diabetic individuals, but only the latter group is overtly glucose intolerant. These observations underscore the important interplay between insulin resistance and insulin secretion. Both abnormalities must be looked at in concert. Insulin resistance alone is, in most instances, insufficient to cause overt glucose intolerance. To observe the development of frank diabetes mellitus, a defect in insulin secretion must be superimposed on insulin resistance. This construct is consistent with the classic overfeeding studies of Sims et al. (88). When healthy subjects received an increased caloric intake and gained 20–30 lbs over 3–5 mo, moderate to severe insulin resistance could be documented via the forearm perfusion technique. Nonetheless, glucose tolerance remained normal in all subjects because the β-cell was able to augment its insulin secretory capacity to offset the impairment in insulin action. Note that the results of insulin-clamp and insulin secretory studies performed in Pima Indians are consistent with the preceding construct (11). Thus, in diabetic Pima Indians, insulin resistance occurs early in the course of the disease and changes little thereafter (Fig. 10 in ref. 11). However, glucose tolerance remains relatively well preserved because of compensatory increases in both the fasting (Fig. 1 in ref. 11) and stimulated (Fig. 6 in ref. 11) plasma insulin responses. However, once the fasting plasma

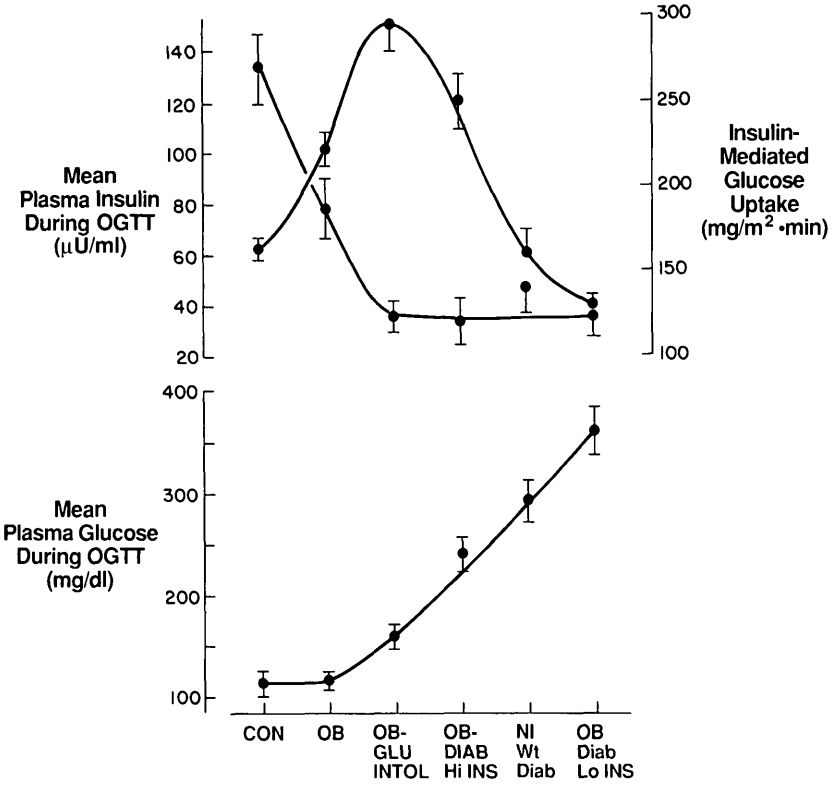


FIG. 7. Summary of plasma glucose (*bottom*) and insulin (*top*) responses during 100-g oral glucose tolerance test (OGTT) and tissue sensitivity to insulin (*top*) in control, obese nondiabetic, obese glucose-intolerant, obese hyperinsulinemic diabetic, normal-weight diabetic, and obese hypoinsulinemic diabetic subjects. See text for details. (Data from ref. 80.)

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glucose concentration exceeded 125 mg/dl, there was a progressive decline in both the fasting and stimulated insulin concentration, which paralleled the rise in fasting plasma glucose concentration.

CELLULAR MECHANISMS OF INSULIN RESISTANCE

To exert its biologic effects on glucose metabolism, insulin must first bind to specific receptors on the cell surface (89,90). After insulin has bound to its receptor, a second messenger is generated (91–93). The precise nature of this second messenger remains unknown. Several candidates have been suggested, but none has gained universal acceptance. Most recently, it has been postulated that the second messenger may be an enzyme complex, which is an integral part of the receptor itself (94,95).

Once it has been generated, the second messenger initiates a series of events that affect the intracellular metabolism of glucose. The initial step involves activation of the glucose transport system, leading to influx of glucose into insulin target tissues, primarily muscle (96,97). The free glucose, which has entered the cell, is subsequently metabolized by a series of enzymatic steps that are under the control of insulin; the most important of these are glycogen synthase (which controls glycogen synthesis) and pyruvate dehydrogenase (which regulates glucose oxidation).

In NIDDM individuals, both receptor and postreceptor defects have been shown to contribute to the insulin resistance. Numerous studies have demonstrated that insulin binding to monocytes and adipocytes from NIDDM patients is reduced by ~20–30% (98–102). This decrease in insulin binding is due to a decrease in the number of insulin receptors without any alteration in the affinity of insulin for its receptor. However, a word of caution should be introduced in interpreting these studies. As previously discussed, the two major target organs for insulin action in vivo are the liver and muscle, and few studies have examined insulin binding to these tissues in humans. Because monocyte and adipocyte binding has been shown to correlate with insulin binding to hepatocytes (103,104) and muscle (105,106) in animals, it has been reasonable to assume that a decrease in insulin binding contributes to the insulin resistance in some NIDDM subjects. However, Arner et al. (107) have challenged this assumption by demonstrating normal insulin binding to hepatic membranes isolated from NIDDM patients.

In contrast, several lines of evidence indicate that decreased insulin binding cannot be the major factor responsible for the defect in insulin action in NIDDM. First, a decrease in insulin-receptor number cannot be demonstrated in approximately one-third to one-half of all NIDDM subjects, especially those with high fasting plasma glucose levels (102,108–113). Second, Lönnroth et al. (114), Olefsky and Reaven (115), and others have been unable to find a correlation between insulin binding and the severity of insulin resistance in their NIDDM patients. In fact, many investigators have not been able to demonstrate any decrease in insulin binding in NIDDM patients (102,108–113). Third, Mandarino et al. (116) and Kolterman et al. (48), when examining the dose-response relationship between insulin-mediated glucose disposal and plasma insulin concentration, have provided evidence to support the existence of a severe postreceptor defect in insulin action. In patients with impaired glucose tolerance and very mild diabetes, the dose-response curve was shifted to the right, but glucose metabolism could be normalized at very high plasma insulin concentrations; this is consistent with a receptor defect, which was indeed documented in isolated adipocytes and monocytes. In contrast, in diabetic patients with moderate to severe fasting hyperglycemia, the dose-response curve was shifted to the right, and insulin-mediated glucose disposal could not be normalized at maximally effective plasma insulin concentrations. These results suggest the presence of a postreceptor defect in insulin action and indeed no decrease in insulin binding could be demonstrated in this group of diabetic patients. Taken together the above results indicate that a postbinding defect in insulin action must be responsible for the observed insulin resistance in diabetic patients with well-established fasting hyperglycemia.

A postbinding defect in insulin action could result from one of three metabolic disturbances: impaired generation of insulin's second messenger, diminished glucose transport into the cell, or a postglucose transport defect in some critical enzyme involved in glucose utilization. Until recently, data concerning the second messenger for insulin action in NIDDM have been lacking. This is largely due to lack of a precise identification of the second messenger for insulin action. It has been suggested that phosphorylation/dephosphorylation of key intracellular proteins may be an important signaling mechanism that couples insulin binding to the intracellular action of insulin (92,117). With regard to this, the α-subunit of the insulin receptor has been shown to contain the major insulin binding domain, whereas the β-subunit expresses insulin-stimulated kinase activity directed toward its own tyrosine residues (89–91). Note that the α-subunit is totally extracellular, whereas the β-subunit is a transmembrane protein (89–91). The most current popular theory suggests that insulin-receptor phosphorylation, with subsequent activation of insulin-receptor tyrosine kinase, may represent an important second messenger for the hormone's action. Freidenberg et al. (118) have shown that insulin-stimulated tyrosine kinase activity of partially purified insulin receptors from adipocytes of obese diabetic patients, but not nondiabetic obese subjects, was decreased by ~50%. Similarly, Comi et al. (119) have shown decreased tyrosine kinase activity in erythrocytes from obese NIDDM individuals. Caro et al. (120) have also documented decreased insulin-stimulated tyrosine kinase activity in muscle and liver (121) from morbidly obese NIDDM subjects but not in weight-matched nondiabetic control subjects. The decrease in tyrosine kinase activity could not be explained by a decrease in insulin binding, which was normal in the latter three studies (119–121). To the extent that the insulin-receptor tyrosine kinase represents an important mediator of insulin action, its decreased activity in NIDDM patients may represent a key intracellular mechanism for insulin resistance.

Once the second messenger for insulin action has been activated, it enhances glucose transport into the cell. This effect is facilitated by the translocation of an intracellular pool of glucose transporters to the cell membrane (96,97). In recent years the glucose transporter has been the focus of great attention, and it has been purified to homogeneity by several groups (for review see ref. 121). Its structure is that of a glycoprotein with a molecular weight of ~55,000. Several

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studies have examined glucose transport activity in NIDDM and have found it to be uniformly diminished (48,108,109, 113,116,122,123). Whether this is due to a decrease in the number of intracellular glucose transport units or to an inability of insulin to translocate them to the cell membrane and cause their activation has yet to be determined. Furthermore, it is unknown whether the defect in insulin-stimulated glucose transport is a primary manifestation of insulin resistance, is secondary to impaired generation of insulin's second messenger, or is the consequence of downregulation of the glucose transport system by hyperglycemia (125; L. Rossetti, R.A.D., G.I. Shulman, S.W. Cushman, and B.B. Kahn, unpublished observations).

Because glucose transport represents an early step in insulin action, if this were the primary cause of the insulin resistance, the two major metabolic pathways of glucose metabolism, i.e., glucose oxidation and glucose storage (nonoxidative glucose disposal), would be expected to be similarly impaired when expressed as a percentage of the total-body glucose disposal rate. In control subjects, after the ingestion of 100 g of glucose, glucose storage accounted for 72 ± 1% of the oral load. In NIDDM patients with fasting plasma glucose concentrations <180 mg/dl, glucose storage represented 60 ± 3% ($P < .05$) of the ingested glucose load and was reduced further to 49 ± 3% ($P < .05$) in diabetic individuals with fasting glucose levels >180 mg/dl. Similar results have been reported by Bogardus and colleagues (11,66,126). Our results, as well as those of Bogardus et al., suggest that defects in enzymatic steps distal to glucose transport may become rate limiting for insulin-mediated glucose disposal in NIDDM subjects with moderately severe to severe glucose intolerance. Whether postglucose transport steps are rate limiting for insulin action in NIDDM individuals with more modest glucose intolerance cannot be resolved by this analysis because glucose oxidation (i.e., pyruvate dehydrogenase activity) and glucose storage (i.e., glycogen synthase activity) could be impaired in parallel.

As stated previously, the two major intracellular fates of glucose are oxidation and storage (nonoxidative disposal). Both of these processes can be measured by indirect calorimetry, and we have shown that both are impaired in NIDDM patients (Fig. 6). Nonoxidative glucose disposal primarily reflects glycogen synthesis (66,84,127–129), because glucose conversion to lipid is negligible (68,69), and little of the glucose taken up by muscle is released as lactate (84). Glycogen synthase is a key insulin-regulated enzyme for glycogen formation (130–132). Bogardus et al. (66) and Mandarino et al. (133) have shown that the ability of insulin to stimulate glycogen synthase in muscle and adipocytes is impaired in NIDDM subjects. The results of the former group are particularly interesting in that the reduced glycogen synthase activity correlated well with both the defects in nonoxidative glucose disposal and muscle glycogen formation. Similar correlations between impaired insulin-mediated glucose disposal, diminished activation of glycogen synthase, and decreased muscle glycogen formation during the insulin clamp have been reported in poorly controlled NIDDM patients (83).

The other major pathway of glucose disposal, i.e., glucose oxidation, is also impaired in NIDDM individuals (Fig. 6). A key regulator of glucose oxidation is pyruvate dehydrogenase, an enzyme regulated by insulin (134). Only one previous study has examined pyruvate dehydrogenase activity in NIDDM patients, and the ability of insulin to stimulate this enzyme complex in adipocytes was found to be impaired (133).

In summary, it appears that both binding and postbinding defects in insulin action contribute to the insulin resistance in NIDDM. However, diminished insulin binding occurs primarily in individuals with impaired glucose tolerance or very mild diabetes. In NIDDM patients with fasting glucose levels >140 mg/dl, diminished insulin binding is unusual, and postbinding defects are primarily responsible for the insulin resistance. Several postbinding defects have been documented, including diminished tyrosine kinase activity, decreased glucose transport, impaired glycogen synthase activity, and reduced pyruvate dehydrogenase stimulation.

ROLE OF LIPID OXIDATION IN PATHOGENESIS OF INSULIN RESISTANCE IN NIDDM

Randle et al. (135,136) more than two decades ago proposed that increased free fatty acid (FFA) oxidation restrains glucose oxidation in muscle by altering the redox potential of the cell and inhibiting several key enzymatic steps within the glycolytic cascade. When FFA is oxidized in excessive amounts, there is an accumulation within the cell of acetyl-CoA—a powerful inhibitor of pyruvate dehydrogenase (135–138). In addition, because of the increased NADH/NAD, the Krebs cycle is slowed, and citrate accumulates. This intermediate is a powerful inhibitor of phosphofructokinase (135,136,139), which leads to product inhibition of the early steps involved in glucose metabolism. Glucose 6-phosphate eventually builds up and inhibits hexokinase, leading to a decrease in glucose transport into the cell. This sequence of events would be expected to impair both glucose oxidation (direct inhibition of pyruvate dehydrogenase and Krebs cycle) and glucose storage, i.e., glycogen formation (secondary to decreased glucose transport). Although not appreciated at the time that Randle published his classic studies, it has become established that increased FFA oxidation can inhibit glycogen synthase activity directly by causing a dissociation of its subunits (140). Thus, an elevated rate of FFA oxidation can reproduce all of the major intracellular abnormalities (decreased glucose transport, decreased glycogen synthase, decreased pyruvate dehydrogenase) observed in the NIDDM subject and could account for the defects in glucose oxidation and glucose storage. Furthermore, we have provided experimental validation in healthy humans that physiologic elevations in the plasma FFA concentration cause a stimulation of FFA oxidation, which in turn inhibits glucose oxidation and glucose storage (141,142). Normal subjects participated in three experimental protocols: *1*) euglycemic insulin (+100-μU/ml) clamp with indirect calorimetry, *2*) insulin clamp plus Intralipid infusion to maintain the plasma FFA concentration at the basal level, and *3*) insulin clamp plus Intralipid infusion to increase the plasma FFA concentration. During the control insulin-clamp study, the plasma FFA concentration declined by ~70%, and this was associated with a similar reduction in lipid oxidation. When the plasma FFA concentration was maintained at the fasting level, the basal rate of lipid oxidation was maintained constant throughout the insulin

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clamp. Finally, when the plasma FFA concentration was increased, lipid oxidation rose. The changes in lipid oxidation were associated with reciprocal dose-related changes in glucose oxidation and glucose storage (nonoxidative glucose disposal). Simply maintaining plasma FFA concentration and lipid oxidation constant at the basal level led to a significant impairment in insulin-mediated rates of glucose oxidation and storage. Elevation of the plasma FFA concentration and lipid oxidation above baseline caused a further impairment in both the oxidative and nonoxidative pathways of glucose metabolism. Lipid infusion has also been shown to impair oral glucose tolerance in normal subjects by inhibiting both glucose oxidation and glucose storage (143,144). These results provide strong evidence that physiologic elevations in plasma FFA levels and lipid oxidation can reproduce the major defects in glucose metabolism that have been described in NIDDM patients.

Over the last decade Felber and colleagues (145–148) have demonstrated that basal plasma FFA levels and lipid oxidation are elevated in human obesity and diabetes and fail to suppress normally after glucose ingestion. The results of our insulin-clamp studies performed in obese nondiabetic and obese diabetic subjects (80; Fig. 6) are in agreement with those of Felber et al. In obese nondiabetic, obese glucose-intolerant, and obese diabetic individuals, both the fasting plasma FFA concentration and basal rate of lipid oxidation (measured by indirect calorimetry) were elevated compared with control subjects and failed to suppress normally during the euglycemic insulin clamp. With multiple stepwise regression analysis, lipid and glucose oxidation were found to be strongly and inversely related to each other during the basal state and during the insulin clamp ($r = -.80$, $P < .001$). The relationship between lipid oxidation and glucose storage during the insulin clamp ($r = .35$, $P < .05$), although present, was much weaker than between lipid and glucose oxidation. Increased postabsorptive plasma FFA concentrations and lipid oxidation, as well as impaired suppression in response to insulin, have been reported by others in nondiabetic obese and obese diabetic individuals (11,149–151). Taken together these results suggest that elevated rates of lipid oxidation contribute to the defects in glucose oxidation and, to a lesser extent, glucose storage in insulin-resistant obese diabetic individuals. In contrast, in normal-weight NIDDM subjects with moderate fasting hyperglycemia, basal rates of lipid oxidation are not increased and suppress normally in response to insulin (A. Golay, R.A.D., E. Ferrannini, D.C. Simonson, D. Thorin, K. Acheson, D. Thiebaud, B. Curchod, E. Jequier, and J.P. Felber, unpublished observations). Thus, it is difficult to invoke increased Randle cycle activity as a cause of the insulin resistance in normal-weight diabetic patients.

An elevated rate of FFA oxidation also has important effects on hepatic glucose metabolism. In vitro studies have demonstrated that FFAs stimulate gluconeogenesis (153, 154). Furthermore, if FFAs are infused into humans under conditions that simulate the diabetic state (142) or into obese insulin-resistant subjects (155), HGP is enhanced. Other investigators have shown that basal plasma FFA levels and increased rates of lipid oxidation are strongly correlated with both the fasting plasma glucose concentration and the basal rate of HGP (11,156). The following sequence might explain the relationship between plasma FFA concentrations, lipid oxidation, and glucose metabolism at the level of the liver: *1*) increased plasma FFA, by mass action, enhances their cellular uptake, which leads to an increase in lipid oxidation that provides the stimulus for activation of the key enzymes involved in the regulation of gluconeogenesis (157); *2*) at the same time, the augmented rate of lipid oxidation provides a continued source of energy (ATP) and substrate to drive gluconeogenesis; *3*) in addition, the uptake of circulating gluconeogenic precursors by the liver is elevated in obesity (158) and NIDDM (60). Note that the inhibitory influence of insulin on gluconeogenesis is much more resistant than its retraining action on glycogenolysis (45).

In summary, the increased fat mass in obese individuals, diabetic and nondiabetic, is associated with an enhanced rate of lipolysis, which leads to an elevation in the plasma FFA concentration (Fig. 8). By a mass-action effect, this increases the cellular uptake of FFA and stimulates lipid oxidation. In muscle the accelerated rate of fat oxidation inhibits insulin-mediated glucose disposal, whereas at the level of the liver it stimulates gluconeogenesis and increases HGP. These observations have important potential therapeutic implications. Thus, drugs that inhibit lipolysis and/or FFA oxidation may prove useful in treating obese NIDDM patients.

PATHOGENESIS OF NIDDM

When NIDDM patients first present to a physician, most have had diabetes for many years, and defects in both insulin action and insulin secretion are present (5). At this stage it is impossible to backtrack in time to define which defect came first in the evolution of their disease. However, it seems likely that in any given diabetic patient, whatever defect, e.g., insulin resistance or impaired insulin secretion, initiates the disturbance in glucose metabolism, it will eventually be followed by the emergence of its counterpart.

Figure 9 represents an attempt to integrate the principal pathogenetic factors responsible for the development of

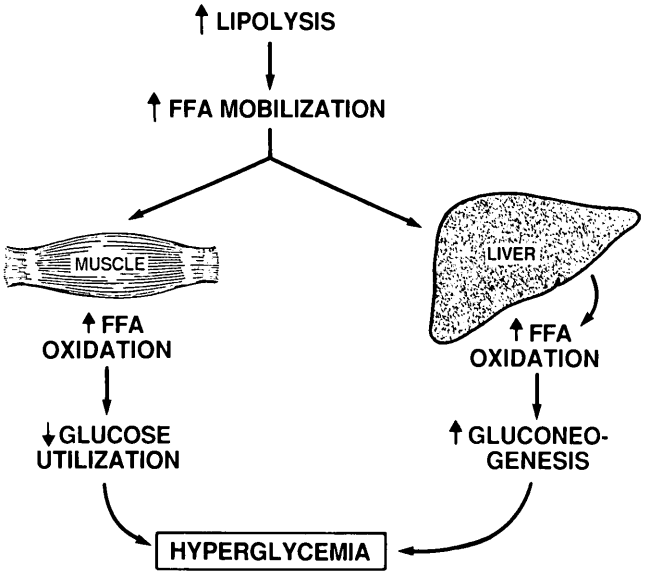


FIG. 8. Effects of elevated plasma free fatty acid (FFA) concentrations and enhanced lipid oxidation on muscle and hepatic glucose metabolism. See text for details.

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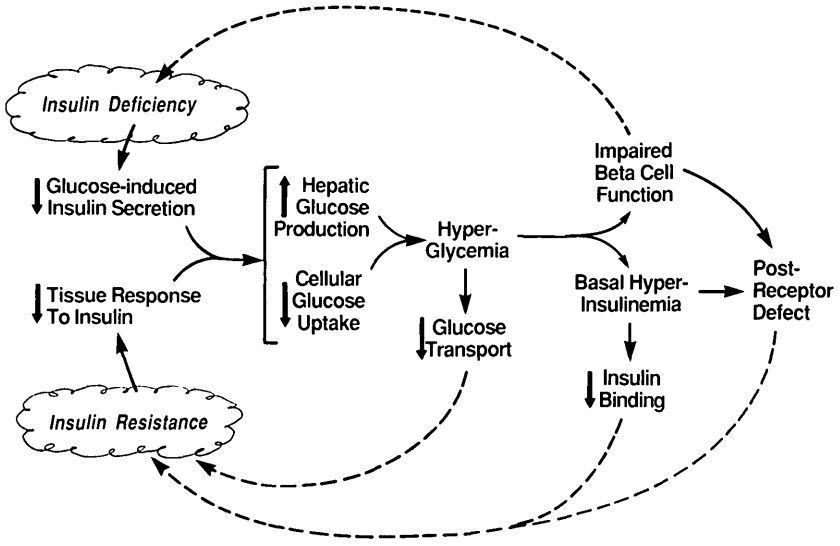


FIG. 9. Pathogenetic sequence of events leading to development of insulin resistance in NIDDM. Note that whether primary defect that initiates glucose intolerance resides in β-cell or in peripheral tissues, development of insulin resistance will eventually ensue or become aggravated, respectively. *Broken arrows* represent positive-feedback loops, resulting in self-perpetuation of primary defects.

NIDDM. It should be emphasized that the schema outlined in this figure is still a working hypothesis that needs to be altered as new data become available. However, this hypothesis has much experimental evidence as its foundation. The major point is that NIDDM is a heterogenous disorder that may be initiated by one of two primary metabolic disturbances: a defect in insulin secretion by the β-cell or a cellular defect in insulin action.

Primary β-cell defect. The plasma insulin response to hyperglycemia in many, but not all, NIDDM subjects is inadequate when viewed in absolute terms (5). However, even in diabetic individuals with normal or increased plasma insulin response, the first phase of insulin secretion is impaired (4–8,159). In fact, when the fasting plasma glucose level exceeds 110–120 mg/dl, the early phase of insulin secretion is uniformly lost. Consequently, there may be an excessive and prolonged rise in plasma glucose concentration after glucose administration (9,10). This postprandial hyperglycemia will present a persistent stimulus to the β-cell, and the total amount of insulin secreted in response to the meal may actually be increased. This postprandial hyperinsulinemia may eventually return the fasting plasma glucose concentration to normal. As the diabetes becomes more severe (as a consequence of the progressive nature of the β-cell defect), the plasma insulin response (although still increased in absolute terms) will become insufficient to return the plasma glucose concentration to its basal level, and fasting hyperglycemia will ensue. Fasting hyperglycemia will present a persistent stimulus to the pancreas to oversecrete insulin throughout the day (6), and fasting hyperinsulinemia will result (5). An important regulator of the number of insulin receptors on insulin target tissues is the hormone concentration itself (160–162). Moreover, most evidence suggests that it is the level of insulin to which cells are chronically exposed throughout the day that regulates the number of insulin receptors on cells. Thus, the persistent basal and glucose-stimulated hyperinsulinemia will lead to a decrease in insulin binding (162) and to the development of insulin resistance. More recently, in vitro (163,164) and in vivo (165–168) studies have demonstrated that prolonged hyperinsulinemia can lead to a downregulation of the intracellular events involved in insulin action.

The pathogenetic sequence of events outlined above is representative of the patient with impaired glucose tolerance (i.e., fasting euglycemia, an impaired but not overtly diabetic glucose tolerance test, fasting hyperinsulinemia, and increased insulin response to glucose) and the patient with early diabetes mellitus (mild fasting hyperglycemia, a diabetic glucose tolerance test, fasting hyperinsulinemia, and normal or increased plasma insulin response to glucose). In such an individual the basal rate of HGP would remain within the normal range because of the suppressive effects of hyperglycemia, whereas the efficiency of tissue glucose uptake would be markedly reduced (Fig. 5).

As the plasma insulin response becomes progressively more impaired, either as a result of the initial defect or pancreatic "exhaustion" (see GLUCOSE TOXICITY), both the early and late phases of insulin secretion will become absolutely deficient. However, the basal plasma insulin level will remain elevated (5) because of the persistent stimulus presented to the β-cell by the fasting hyperglycemia (6). With the onset of insulinopenia, whether relative or absolute (Fig. 1), marked fasting hyperglycemia will ensue due to excessive glucose production by the liver (Fig. 4) and a further reduction in tissue glucose clearance (Fig. 5). The clinical picture will be that of a typical NIDDM patient.

Another important pathogenetic disturbance that emerges with the development of insulinopenia is a postreceptor defect in insulin action. Many of the intracellular events involved in glucose metabolism depend on the surge of insulin that occurs three to four times per day in response to nutrient ingestion. When the insulin response becomes deficient, the activity of the glucose transport system becomes severely impaired (122,169–172), and several intracellular enzymatic steps involved in glucose metabolism are depressed (83, 131,173,174). Additionally, when severe insulinopenia ensues, plasma FFA levels and fatty acid oxidation will increase, further contributing to the defects in intracellular glucose disposal (Fig. 8).

In summary, the sequence of events, starting with a defect in insulin secretion and leading to the emergence of insulin resistance, appears to be plausible on the basis of available information. Furthermore, this pathogenetic sequence explains the development of receptor (predominant in individ-

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uals with impaired glucose intolerance and mild diabetes mellitus) and postreceptor (predominant in patients with moderate to severe diabetes mellitus) defects in insulin action. The sequence of events outlined above has received validation by experimental observations in dogs (175–177), rats (125), and humans (178).

Primary cellular (muscle) defect. The plasma insulin response to oral and intravenous glucose has been shown to be increased in absolute terms in normal-weight individuals with impaired glucose tolerance, in many normal-weight patients with mild diabetes mellitus, and especially in obese subjects (nondiabetic and diabetic; Table 2; 5,158). Furthermore, it should be remembered that most (70–80%) NIDDM individuals in Western countries are overweight. The disturbance in glucose metabolism in these three subsets of patients is probably initiated by a primary cellular defect (receptor or postreceptor) in insulin action. Because of the compensatory increase in insulin secretion, initially the insulin resistance by itself is not sufficient to impair glucose uptake by muscle tissue and to cause an increase in HGP. However, as the insulin resistance progresses, muscle glucose uptake becomes further impaired, and eventually the fasting plasma glucose concentration rises; HGP remains within the normal range because of the hyperinsulinemia. The development of hyperglycemia further stimulates β-cell secretion, and the resultant hyperinsulinemia causes a downregulation of both insulin-receptor number and intracellular events involved in insulin action, thus exacerbating the insulin resistance. In some individuals the persistent stimulus to the β-cell to oversecrete insulin will lead to a demise in β-cell function. With regard to this, recent studies have suggested that hyperglycemia per se may contribute to the defect in insulin secretion (see GLUCOSE TOXICITY). The resultant insulinopenia will lead to the emergence, or exacerbation, of postreceptor defects in insulin action. In addition, there is mounting evidence that hyperglycemia can downregulate the glucose transport system as well as other intracellular events involved in insulin action. This pathogenetic sequence could explain the clinical features observed in most NIDDM patients, particularly those who are overweight. Insofar as the cellular defect is generalized, both hepatic and peripheral tissues (and possibly the β-cells themselves) would manifest the insulin resistance, and the numerous metabolic alterations characteristic of the diabetic state could be related to a single primary defect.

The sequence of events detailed above has received considerable validation in humans. Thus, if normal-weight subjects are overfed and their ideal body weight increases by as little as 15–20%, moderate to severe peripheral insulin resistance develops (88), and their liver becomes refractory to the suppressive action of insulin on HGP (175). Because the subjects studied by Sims et al. (88) were followed for only a relatively short period, it was not possible to determine whether the insulin resistance eventually resulted in any impairment of β-cell function. Note, however, that most obese individuals maintain perfectly normal glucose tolerance. Therefore, it is likely that overt NIDDM will develop only in obese individuals with a concomitant defect in β-cell secretion of insulin. The reverse experiment to the overfeeding studies has been performed and provides strong support for the preceding sequence of events. Thus, when obese insulinopenic diabetic subjects lose weight, fasting hyperglycemia is improved, and tissue sensitivity to insulin is enhanced (179–182). This gives the exhausted β-cell a chance to rest, and after several weeks to months, a marked improvement in insulin secretion ensues (183–188). If ideal body weight can be achieved and maintained, glucose tolerance can be markedly improved or even normalized in most obese NIDDM patients (182).

Other clinical evidence supports the primacy of insulin resistance in the pathogenesis of NIDDM. First, if the offspring of two NIDDM parents are followed sequentially before they develop overt diabetes mellitus, the earliest detectable abnormality is a decrease in tissue sensitivity to insulin; neither the early or late phase of insulin secretion is impaired (189). Second, aggressive insulin therapy with normalization of the plasma glucose profile and glycosylated hemoglobin levels has failed to reverse (190–192) or only minimally improves (188,192–194) the insulin resistance in NIDDM. However, it could be argued that in these studies, insulin therapy was instituted late in the course of NIDDM after fixed irreversible metabolic disturbances had developed (187,189–193). Alternatively, the resultant hyperinsulinemia, which was required to normalize glucose tolerance, may itself have led to the emergence of binding and postbinding defects in insulin action (160–168). Third, in the Pima Indians, NIDDM is inherited as an autosomal dominant (194a). In this population NIDDM is universally associated with insulin resistance, and recent observations indicate that there is a single gene for insulin resistance with a codominant mode of inheritance (C. Bogardus, unpublished observations).

In summary, the much and seemingly disparate evidence concerning the pathogenesis of NIDDM can be distilled down to the interaction of the two classic defects, insulin deficiency and insulin resistance (Fig. 9). Either can be primary and is sufficient to induce hyperglycemia. On clinical and epidemiological grounds, it would appear that the two defects may prevail in different subsets of the diabetic population, possibly by virtue of stronger penetrance of their genetic determinant. Until some specific marker for insulin deficiency or insulin resistance is found, the problem of the etiology of each form of diabetes will remain unsolved. However, the key concept is that the two fundamental defects are intimately related to one another in the pathogenesis of diabetes mellitus. As outlined in Fig. 9, each of them can lead to the other secondarily and by more than one mechanism. We view this as a conceptional advancement over previous controversies on the primacy of insulin deficiency or insulin resistance and our focus is therefore on the modes and sites of their interaction. Below I discuss how β-cell exhaustion and receptor/postreceptor downregulation can aggravate insulin deficiency and insulin resistance, respectively, through the common step of prolonged hyperglycemia, whether induced by one or the other primary defect (Fig. 9).

GLUCOSE TOXICITY: AN EMERGING CONCEPT

In the last decade the importance of insulin resistance in the pathogenesis of NIDDM has become well established. However, the concept that glucose per se (i.e., hyperglycemia) may be a cellular toxin is less well appreciated. Nonetheless,

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this is an important concept in understanding the pathogenesis of NIDDM. Although still a hypothesis, significant data have accumulated to support the concept of glucose toxicity. This hypothesis can be stated

> Chronic hyperglycemia may cause a generalized desensitization of all cells in the body through downregulation of the glucose transport system. In muscle and adipocytes this would be reflected by a defect in insulin action, whereas at the level of the β-cell, this would be manifested by an impairment in insulin secretion.

Unger and Grundy (195) were the first to suggest that hyperglycemia may be partly responsible for the impairment in insulin-mediated glucose disposal in both NIDDM and insulin-dependent diabetic (IDDM) subjects. We provided the first experimental proof of this hypothesis in an animal model of NIDDM (125). Four groups of chronically catheterized, awake, unstressed rats were studied: group 1, sham operated controls; group 2, 90% surgically pancreatectomized rats; group 3, 90% pancreatectomized rats treated with phlorizin (an inhibitor of renal tubular glucose transport) to normalize the plasma glucose profile; and group 4, diabetic rats treated with phlorizin for 6 wk and subsequently restudied 2 wk after discontinuation of the phlorizin. Groups 1–3 were studied 6 wk after pancreatectomy or sham pancreatectomy.

The pancreatectomized rats had mild fasting hyperglycemia and a markedly abnormal meal tolerance test compared with the control rats. Insulin secretion was markedly impaired in the diabetic group. After phlorizin treatment, the fasting plasma glucose concentration and the postmeal glucose profile were normalized, without any change in plasma insulin levels. When the phlorizin was discontinued glucose intolerance returned, and the plasma insulin response remained markedly deficient.

If hyperglycemia is an important regulator of insulin action in vivo, the diabetic animals should be insulin resistant, and phlorizin would be expected to improve the defect in insulin-mediated glucose disposal. As can be seen in Fig. 10, this is precisely what occurred. Not only was insulin sensitivity improved in phlorizin-treated diabetic rats, it was completely returned to normal. As further proof of the deleterious effect of hyperglycemia on insulin-mediated glucose disposal, discontinuation of phlorizin therapy led to a return of the insulin resistance.

Furthermore, if the theory of glucose toxicity is correct, a downregulation of the glucose transport system in insulin target tissues should be found. To examine this question, 3-*O*-methylglucose transport activity was quantitated in adipocytes isolated from groups 1–3 (124). Both basal and insulin-stimulated glucose transport were significantly reduced in diabetic compared with control rats, and both returned to normal after phlorizin treatment. Consistent with these observations, Matthaei et al. (196) have shown that chronic hyperglycemia in vivo caused a 40–50% decrease in the number of glucose transport units in the plasma membranes and microsomes isolated from brain microvessels of diabetic rats. Likewise, incubation of rat skeletal muscle (197) and adipocytes (198) in vitro with high-medium glucose concentrations is associated with a reduction in glucose transport activity and insulin resistance. Conversely, incubation of various cell types in the absence of glucose leads to an increase in the number of glucose transport units and their activity (199–202).

In summary, the results of these animal studies have provided convincing evidence that chronic hyperglycemia can lead to the development of insulin resistance both in vivo and in vitro through downregulation of the glucose transport system.

The other half of the glucose toxicity story states that chronic hyperglycemia will cause a densitization of the β-cell to an acute glycemic stimulus. Much support for this thesis is available from in vitro studies. Bonner-Weir et al. (203) and Kergoat et al. (204) were among the first to demonstrate that chronic sustained hyperglycemia, in the presence of a reduced β-cell mass, can lead to the development of a defect in insulin secretion. With the 90% pancreatectomized (203) and neonatal streptozocin-induced (204) diabetic rat models, both groups showed that the ability of perfused pancreatic tissue in vitro to respond to glucose was markedly impaired, although the insulin secretory response to nonglucose stimuli, e.g., arginine (203,204) and leucine (205), remained intact. Similarly, much evidence has accumulated that the β-cells of human NIDDM subjects have acquired a specific defect in their ability to respond to glucose (7,206). However, no in vivo study has yet demonstrated that chronic hyperglycemia can lead to the development of a defect in insulin secretion in otherwise healthy β-cells or whether restoration of normoglycemia can correct the impairment in insulin secretion. To examine these questions, we performed hyperglycemic clamp studies in the following three groups of rats: sham operated controls, 90% partially pancreatectomized rats, and partially pancreatectomized rats treated with phlorizin to normalize the plasma glucose profile (208). During the study, the plasma glucose concentration was acutely raised and maintained at 125 mg/dl above the baseline level for 120 min. This allowed us

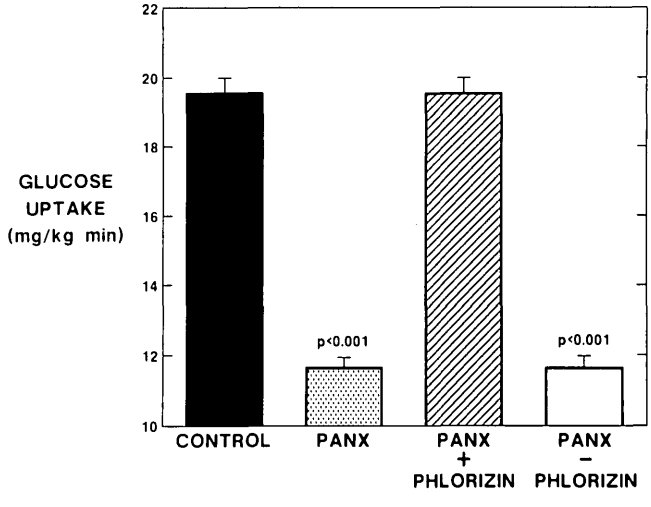


FIG. 10. Total-body glucose uptake during euglycemic insulin (90 μU/ml)-clamp studies performed in control, partially pancreatectomized diabetic rats (PANX), and PANX rats treated with phlorizin. Fourth group of phlorizin-treated PANX rats was restudied after cessation of phlorizin. PANX developed moderate degree of insulin resistance that was completely corrected with phlorizin treatment and returned after phlorizin was stopped. (Data from ref. 125.)

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to quantitate both the early (0- to 10-min) and late (10- to 120-min) phases of insulin secretion. Diabetic animals showed a dramatic reduction in both the early and late phases of insulin release, and these abnormalities were completely corrected by phlorizin treatment. We also tested the response to arginine. In diabetic rats the ability of arginine to stimulate insulin secretion under basal conditions of glycemia was markedly potentiated, and this potentiating effect was reversed by normalization of the plasma glucose profile by phlorizin. These results provide evidence that the β-cells from chronically hyperglycemic animals have acquired a specific "blindness" to glucose. The normal or potentiated response to arginine and other nonglucose secretagogues also explains why the plasma insulin profile throughout the day (when ingesting mixed meals) is not diminished in NIDDM individuals (2).

If the concept of glucose toxicity is correct, normalization of the plasma glucose profile, regardless of the means, should lead to an improvement in insulin secretion in NIDDM subjects. In fact, this study was performed ~8 yr ago by Kosaka et al. (188). Obese NIDDM patients were treated with one of three regimens: diet (weight loss), insulin, or sulfonylureas to improve glycemic control. Each of the therapeutic interventions led to a marked and similar improvement in glycemic control. After successful treatment, the plasma insulin response during a 100-g OGTT showed a twofold increase. Similar improvements in insulin secretion have been demonstrated by others after weight loss (181–187), insulin therapy (73,191,196), and sulfonylurea administration (38,73,209,210).

In summary, there appears to be abundant evidence from both animal and human studies that sustained chronic hyperglycemia can lead to the development of defects in both insulin secretion and insulin action. Furthermore, it appears that these deleterious effects of hyperglycemia are mediated by downregulation of the glucose transport system. Recognition of the important pathogenetic role of hyperglycemia per se in the evolution of NIDDM has important therapeutic implications. Thus, in the poorly controlled diabetic with fasting plasma glucose levels >200 mg/dl it seems reasonable to institute a short course of intensive insulin therapy (4–6 wk) or severe caloric restriction (10–14 days under close supervision by a physician) to restore normoglycemia and reverse the deleterious effects of hyperglycemia on cell function.

CONCLUSION

In conclusion, it has become clear over the past few years that any simplistic view about the pathogenesis of NIDDM in any given patient, such as invoking a single primary defect, is probably misleading. At our current state of knowledge, it is not possible to define, in a given diabetic individual, which is the primary defect, i.e., insulin resistance or impaired insulin secretion. Four new, additional pieces of information must be considered. *1*) Insulin deficiency and insulin resistance usually coexist in overt NIDDM (fasting plasma glucose >140 mg/dl) because they are interrelated at multiple levels (Fig. 9). *2*) In the earliest stages of the evolution of NIDDM (fasting glucose <140 mg/dl), insulin resistance, involving both hepatic and peripheral (muscle) tissues, is a universal finding; however, because the dose-response curve relating the plasma insulin concentration to suppression of HGP lies far to the left of that for insulin-stimulated tissue glucose uptake, HGP remains within the normal range, and decreased efficiency of tissue glucose uptake is primarily responsible for early fasting hyperglycemia. *3*) At the cellular level, multiple defects (decreased glucose transport, impaired activation of tyrosine kinase, and diminished glycogen synthase and pyruvate dehydrogenase activity) contribute to the abnormalities in glucose oxidation and nonoxidative glucose disposal in frankly diabetic individuals. *4*) Hyperglycemia is not only the principal metabolic derangement of diabetes but also a pathogenetic factor in its own right. Once it develops, either from insulin deficiency or from insulin resistance, hyperglycemia will exacerbate both defects, thereby closing a pathologic feedback loop. If hyperglycemia is an effect and a cause of diabetes concurrently, then NIDDM can be viewed as a self-perpetuating disease akin to autoimmune disorders.

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